

**AN INTERNSHIP REPORT
ON
“JOB DHUNDO”**

**Submitted to
SAVITRIBAI PHULE PUNE UNIVERSITY**

In Partial Fulfilment of the Requirement for the Award of

**BACHELOR’S DEGREE IN
COMPUTER ENGINEERING**

BY

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**UNDER THE GUIDANCE OF
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AFFILIATED TO



SAVITRIBAI PHULE PUNE UNIVERSITY

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CERTIFICATE

This is certify that the project entitled

“JOB DHUNDO“

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is a record of bonafide work carried out by them, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Engineering (Computer Engineering) at Trinity Academy of Engineering, Pune under Savitribai Phule Pune University. This work is done during year 2023-2024, under our guidance.

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ABSTRACT

The **Job Dhundo** project is a web-based job portal designed to connect job seekers with employers in a simple, efficient, and user-friendly manner. The primary objective of this system is to streamline the job searching and recruitment process by providing a centralized platform where users can explore job opportunities and recruiters can find suitable candidates. The platform allows job seekers to create profiles, upload resumes, search for jobs based on skills, location, and preferences, and apply directly through the system. On the other hand, employers can register, post job openings, manage applications, and shortlist candidates based on their requirements. The system also includes features like real-time notifications, advanced filtering, and secure authentication to enhance user experience.

Keywords: Job Portal, Online Recruitment System, Job Search Platform ,Resume Management, Candidate Matching, Employer Dashboard, Real-Time Notifications, Full Stack Web Application, Node.js & Express.js, MongoDB Database, User Authentication

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1. About Topic

JOB DHUNDO : A platform to find a secure job for job seekers.

1.1. **Domain** : Web Development, involves html bootstrap, and for backend Django framework.

1.2. **Problem Statement** : The project aims to help students and job seekers to find the relevant job opportunities across multiple platforms and provide a personalized job recommendation .

1.3. **Aim** : The aim of this project is to develop an intelligent job and internship search platform that provides personalized notifications and domain specific recommendation, making job hunting efficient and hassle-free.

1.4. Objectives:

- To Develop a user-friendly web and mobile platform for job seekers
- To Provide real time notifications for job openings in chosen domains
- To simplify the job search processes by reducing irrelevant listings
- To enhance employment opportunities for students and professionals in the Tech industry.
- To integrate a user authentication system.

2. Introduction

2.1. **Need:** In today's fast-paced digital world, finding suitable job opportunities and hiring the right candidates has become a challenging task. Traditional job search methods are time-consuming, inefficient, and often lack proper communication between job seekers and employers. Many candidates struggle to find jobs that match their skills, while recruiters face difficulty in identifying qualified applicants from a large pool.

2.2. **Motivation:** The motivation for developing Job Dhundo comes from the increasing demand for a smart and user-friendly job portal that bridges the gap between job seekers and recruiters. Many existing platforms are either too complex, lack personalization, or do not effectively match candidates with relevant opportunities. As a full-stack developer, building this project also provides practical exposure to real-world problem solving using technologies like JavaScript, Node.js, and databases. The goal is to create a system that not only improves the job search experience but also demonstrates modern web development skills, scalability, and efficient data handling.

2.3. **Basic Concepts:** The Job Dhundo project is based on several fundamental concepts of web development and system design:

- Frontend Development: Designing interactive user interfaces using HTML, CSS, and JavaScript (or React.js)
- Backend Development: Handling server-side logic using Node.js and Express.js
- Database Management: Storing user data, job listings, and applications using MongoDB
- User Authentication: Secure login and registration using sessions or JWT (JSON Web Tokens)
- RESTful APIs: Enabling communication between frontend and backend
- Search & Filtering: Allowing users to find jobs based on skills, location, and preferences
- Cloud Deployment (Optional): Hosting the application for real-world accessibility

3. Literature Survey

3.1. Paper 1: Job Recommender Systems and Recruitment Automation

The literature on job recommender systems highlights their importance in modern recruitment processes. These systems are designed to match job seekers with suitable job opportunities by analyzing large volumes of data such as resumes, job descriptions, and user preferences. Traditional recruitment processes are time-consuming and inefficient, often taking weeks or months to find the right candidate. Job recommender systems aim to automate this process and improve efficiency using advanced technologies.

A key aspect discussed in the literature is the use of Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) in recruitment systems. These technologies help in extracting meaningful information from resumes and job postings, such as skills, qualifications, and experience. By analyzing this data, systems can identify the most suitable candidates for specific job roles and reduce manual effort in hiring.

The literature also emphasizes different types of recommendation techniques, including content-based filtering, collaborative filtering, and hybrid approaches. These methods help improve matching accuracy by considering both job requirements and user preferences. However, challenges such as data inconsistency, bias in algorithms, and lack of personalization still exist in many systems.

Overall, the research concludes that job recommender systems significantly enhance recruitment efficiency by automating matching and improving decision-making. However, further improvements are required to handle complex data and provide more personalized job recommendations.

3.2. Paper 2: AI-Based Personalized Job Recommendation System

This paper focuses on AI-based personalized job search systems that use real-time data and skill-based matching techniques. The system analyzes user profiles, including skills, experience, and preferences, to recommend relevant job opportunities. Unlike traditional systems, it provides personalized results tailored to individual users, improving job search efficiency.

The methodology involves collecting and processing large datasets from job portals and user profiles. Machine learning algorithms are used to identify patterns and relationships between job requirements and candidate skills. These models continuously improve by learning from user interactions, making the system more accurate over time.

Another important aspect discussed is real-time data integration. The system updates job recommendations dynamically based on new job postings and user activity. This ensures that users receive the most relevant and up-to-date job opportunities. Real-time processing also enhances user engagement and satisfaction.

Overall, the literature highlights that AI-based recommendation systems improve job search experiences by providing accurate, personalized, and timely suggestions. However, challenges such as data privacy and model accuracy still need to be addressed.

3.3. **Paper 3: Resume-Based Job Matching System (RésuméMatcher)**

The literature on resume-based job matching systems focuses on improving the relevance of job search results. Traditional job portals rely on keyword-based search, which often produces a large number of irrelevant results. This creates difficulty for job seekers in identifying suitable jobs.

To overcome this limitation, the RésuméMatcher system introduces a model-based approach where both resumes and job descriptions are converted into structured representations. The system calculates similarity scores between candidate profiles and job requirements to determine the best matches. This approach ensures that search results are more accurate and meaningful.

The literature also discusses the concept of personalized job search. Unlike traditional systems where all users receive similar results, this system provides customized results based on individual profiles. This improves the efficiency of job searching and increases the chances of successful job applications.

Overall, the research concludes that resume-based matching systems significantly improve job search quality by providing personalized and relevant results. However, the system depends heavily on accurate data extraction and modeling techniques.

3.4. Paper 4: AI-Integrated Recruitment and Intelligent Job Matching

This paper discusses modern AI-integrated job portals that combine resume analysis, intelligent matching, and automated application tracking. These systems use AI to analyze resumes and match candidates with suitable job roles based on their skills and experience.

A major contribution of this research is the automation of the recruitment process. The system reduces manual effort by automatically screening resumes and recommending candidates to employers. It also provides features such as real-time job alerts and automated application tracking, improving the overall efficiency of the system.

The literature further highlights the role of intelligent matching algorithms in improving recruitment accuracy. These algorithms analyze multiple factors such as skills, experience, and job requirements to provide better matching results. This leads to improved hiring decisions and reduced recruitment time.

Overall, the study concludes that AI-integrated job portals enhance recruitment efficiency, improve user experience, and provide scalable solutions for modern hiring systems. However, issues such as algorithm bias and fairness need further research.

3.5. Summary:

The reviewed literature highlights the growing importance of online job portals and AI-based recruitment systems in modern hiring processes. Traditional job search methods are inefficient and time-consuming, leading to the development of automated job recommender systems that improve matching accuracy and reduce manual effort. These systems use advanced technologies such as machine learning, natural language processing, and data analytics to analyze resumes and job descriptions effectively.

The literature also emphasizes the role of personalization and real-time updates in enhancing user experience. AI-based systems provide tailored job recommendations based on user profiles and continuously update results based on new data. Resume-based matching systems further improve accuracy by using similarity models instead of simple keyword matching.

Additionally, modern recruitment platforms integrate automation features such as resume screening, application tracking, and intelligent candidate selection. These features significantly improve efficiency for both job seekers and employers. However, challenges such as data privacy, algorithm bias, and system scalability remain important areas for future research.

Overall, the literature supports the development of an intelligent and efficient job portal system like **Job Dhundo**, which aims to improve the recruitment process through better matching, user experience, and real-time interaction.

4. Proposed System

4.1. Proposed System in Detail:

- The proposed system is a web-based job portal designed to connect job seekers and employers efficiently. It provides features such as job searching, application management, and real-time updates.
- 1. User Authentication Module**
Handles registration and login for job seekers and employers.
 - 2. Job Seeker Module**
Allows users to create profiles, upload resumes, search jobs, and apply.
 - 3. Employer Module**
Enables recruiters to post jobs, view applications, and shortlist candidates.
 - 4. Job Management Module**
Manages job postings, updates, and deletions.
 - 5. Search & Filter Module**
Provides filtering based on skills, location, and job type.
 - 6. Application Tracking Module**
Tracks applied jobs and their status.
 - 7. Notification Module**
Sends updates about job postings and application status.
 - 8. Notification Module**
Sends updates about job postings and application status.

4.2. Methodology

The proposed system follows a modular development methodology combining web development principles with AI-based processing.

Methodology 1: Job Portal Design

This methodology focuses on designing a user-friendly and interactive interface for both job seekers and employers. It involves analyzing user requirements and creating an intuitive layout that allows easy navigation across different features such as job search, profile management, and application tracking. The design ensures responsiveness across devices and emphasizes usability, accessibility, and a clean UI/UX structure. Special attention is given to simplifying complex processes like job filtering and application submission, ensuring that users can interact with the system efficiently without confusion.

Methodology 2: Backend Development

This methodology involves implementing the server-side logic of the system using technologies like Node.js and Express.js. It includes handling user requests, processing data, and managing communication between the frontend and database through RESTful APIs. The system is designed to ensure secure authentication, efficient data processing, and smooth execution of operations such as job posting, application submission, and user management. Error handling, validation, and scalability are also considered to ensure that the backend can support multiple users and large volumes of data without performance issues.

Methodology 3: Database Handling

This methodology focuses on efficient storage, retrieval, and management of data using MongoDB. It involves designing database schemas for users, job listings, and applications to ensure data consistency and integrity. The system uses structured collections to store information and implements queries to fetch relevant data quickly based on user actions. Indexing and optimization techniques are applied to improve performance, especially during search and filtering operations. Security measures are also considered to protect sensitive user data and maintain privacy.

Methodology 4: Real-Time Features

This methodology focuses on enhancing user interaction by implementing real-time features such as notifications and updates. Technologies like asynchronous requests (AJAX) are used to ensure that users receive instant updates without reloading the page. This includes notifying job seekers about application status and informing employers about new applicants. The system ensures quick data synchronization between client and server, improving responsiveness and overall user experience. Real-time functionality plays a key role in making the platform dynamic, interactive, and efficient.

4.3. Algorithm

Algorithm 1: Job Search

- Step 1: User enters search criteria
Step 2: System queries database
Step 3: Filter results
Step 4: Display jobs

Algorithm 2: Job Application

- Step 1: User selects job
Step 2: Submit application
Step 3: Store in database
Step 4: Notify employer

Algorithm 3: Candidate Selection

- Step 1: Employer views applications
Step 2: Filter candidates
Step 3: Shortlist candidates
Step 4: Notify selected users

5. Implementation

1.Login Page: Users log in securely using credentials.

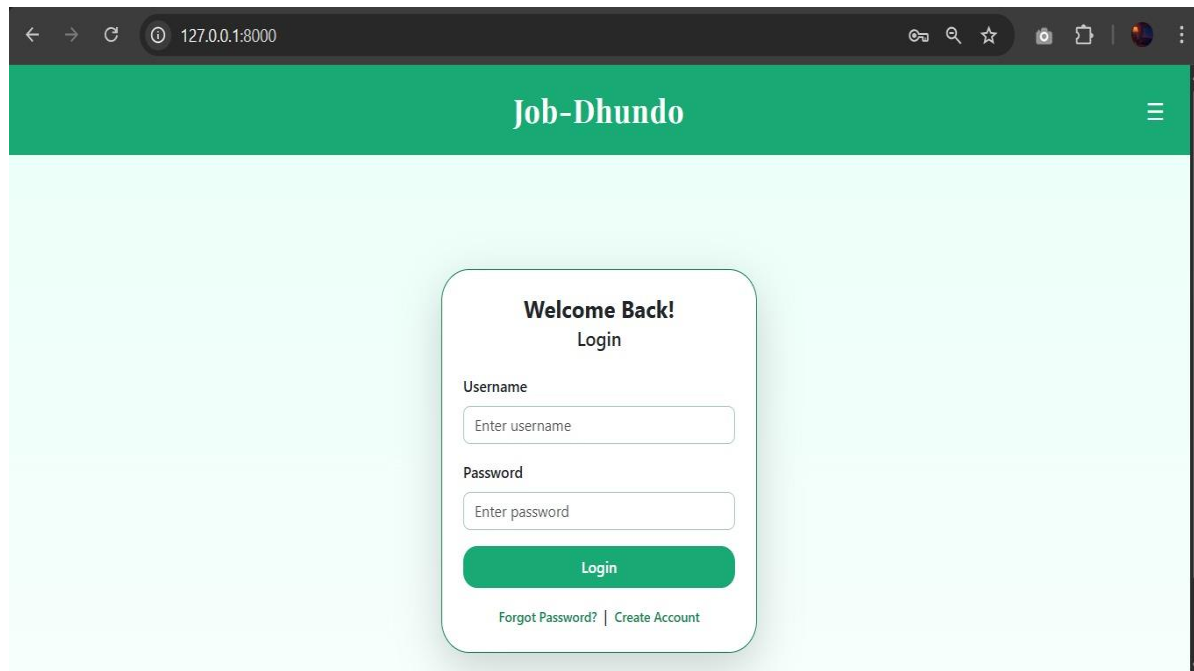
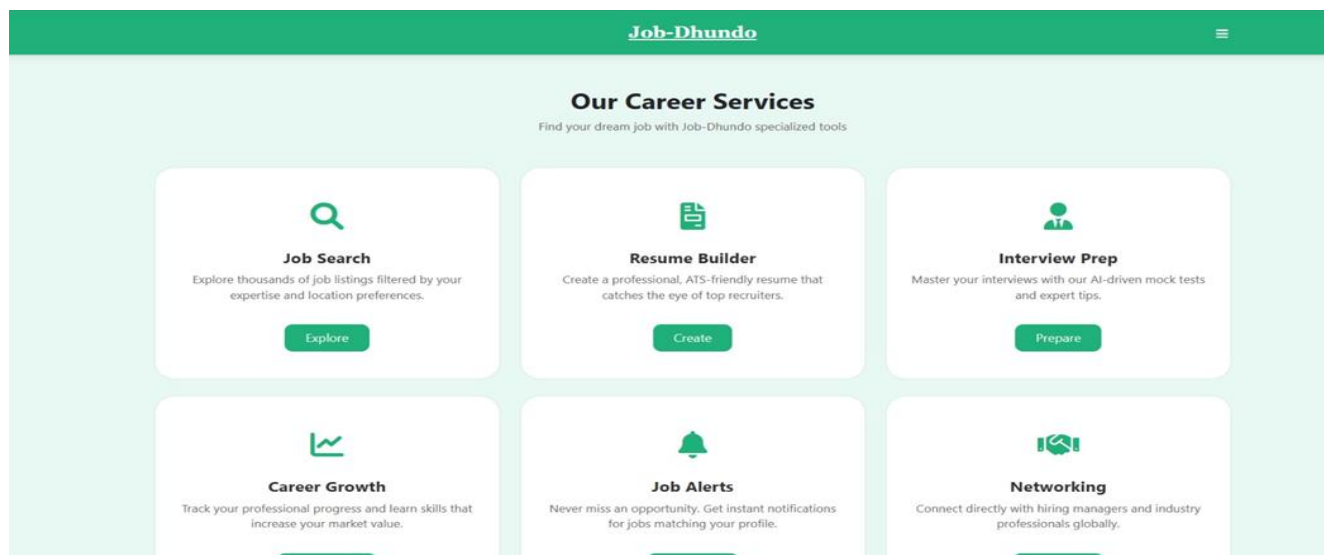


Fig 2

2.Home Feed / Dashboard: Displays jobs, profile, and options..



3. **Job Search Page:** Users search and filter jobs.



Fig 4

4. **Messages page :** Users apply and track status.

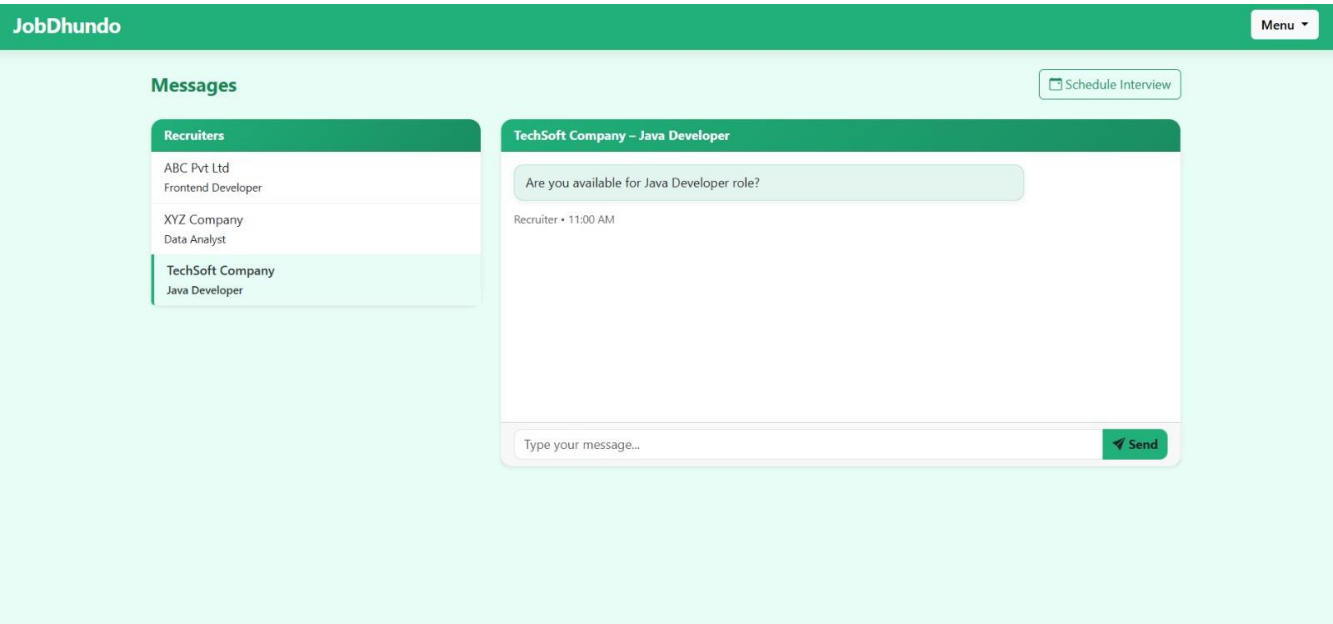


Fig 5

5. Resume Dashboard: Manage job postings and candidates.

Job-Dhundo

Contact Details

Summary

Experience

Education

Skills

Profile Strength

20% Completed

Contact Information

First Name

e.g. Rahul

Last Name

e.g. Sharma

Profile Photo

Upload image

Email Address

rahul.sharma@example.com

Phone Number

+91 98765 43210

Continue to Summary

Auto-saving...

Your data is private & secure

5. Conclusion

The **Job Dhundo** system successfully provides an efficient platform for connecting job seekers and employers. It simplifies the recruitment process through features like job search, application tracking, and real-time updates. The system is scalable, user-friendly, and suitable for real-world implementation. It serves as a strong foundation for developing advanced recruitment platforms in the future.

Furthermore, the **Job Dhundo** system demonstrates the practical application of full-stack web development concepts in solving real-world recruitment challenges. By integrating features such as efficient job search, secure authentication, and real-time communication, the platform enhances both usability and performance. The project also highlights the importance of scalable architecture and modular design in building modern web applications. Overall, it not only simplifies the hiring process but also provides a foundation for future enhancements such as AI-based recommendations and advanced analytics, making it a robust and future-ready

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